Jago Descends

by Louise Thomas

In October 1998, German and New Zealand scientists investigated a cold, lightless world where no human had ever been. The mysterious place they explored was the very bottom of Lake Taupo. The water pressure at the lake floor would quickly crush the life out of a person – if the cold didn't kill them first. To survive more than a few moments on the lake bed, the scientists would need to take a life-giving environment with them. And so they travelled inside a little, two-person submarine called *Jago*.

Nobody had ever explored the lake bed before.

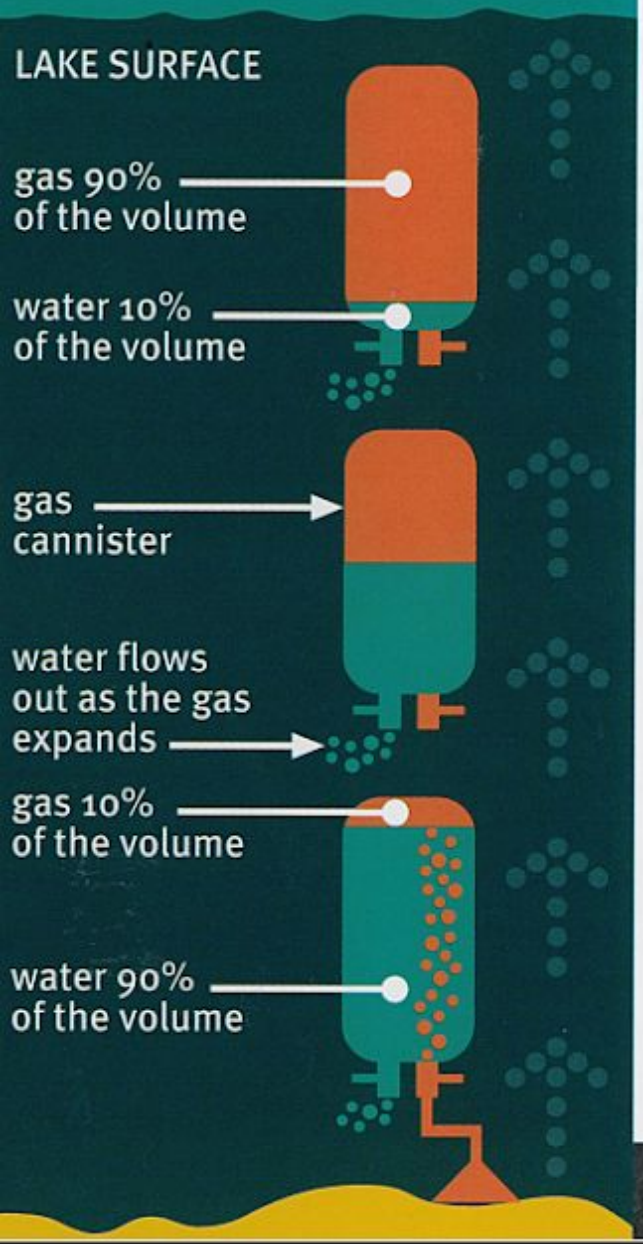
STANHOPE ROAD
SCHOOL

Jago was brought all the way from Germany because we didn't have a research submarine in New Zealand. *Jago* arrived in Wellington on 15 October and was taken by truck to Lake Taupo. Nobody had ever explored the lake bed before. Down, down, down went the small bubble of life. Finally, the scientists arrived at the lake bed. At a depth of 188 metres, they were a very long way from the airy world above.

They quickly set to work collecting rocks, mud, gas bubbles, and hot water from the bottom of the lake. *Jago* has a pair of robotic arms, each of which ends in a three-fingered hand. The mud is made up of very fine rock grains, soil, and sand that wash off the land and sink to the bottom of the lake. The water they took was hot because it comes from underground vents in the lake bed. This water is heated by rock deep in the Earth's crust. The heat is transferred to the water in the same way that heat from a stove element is transferred went the small bubble of life.

Down, down, down

The scientists took samples of gas (mainly carbon dioxide) that was found bubbling up from vents. At depth, the gas is under great pressure. As *Jago* returned to the surface, the pressure decreased, and so the gas expanded to about ten times the volume it had occupied at the lake bed.

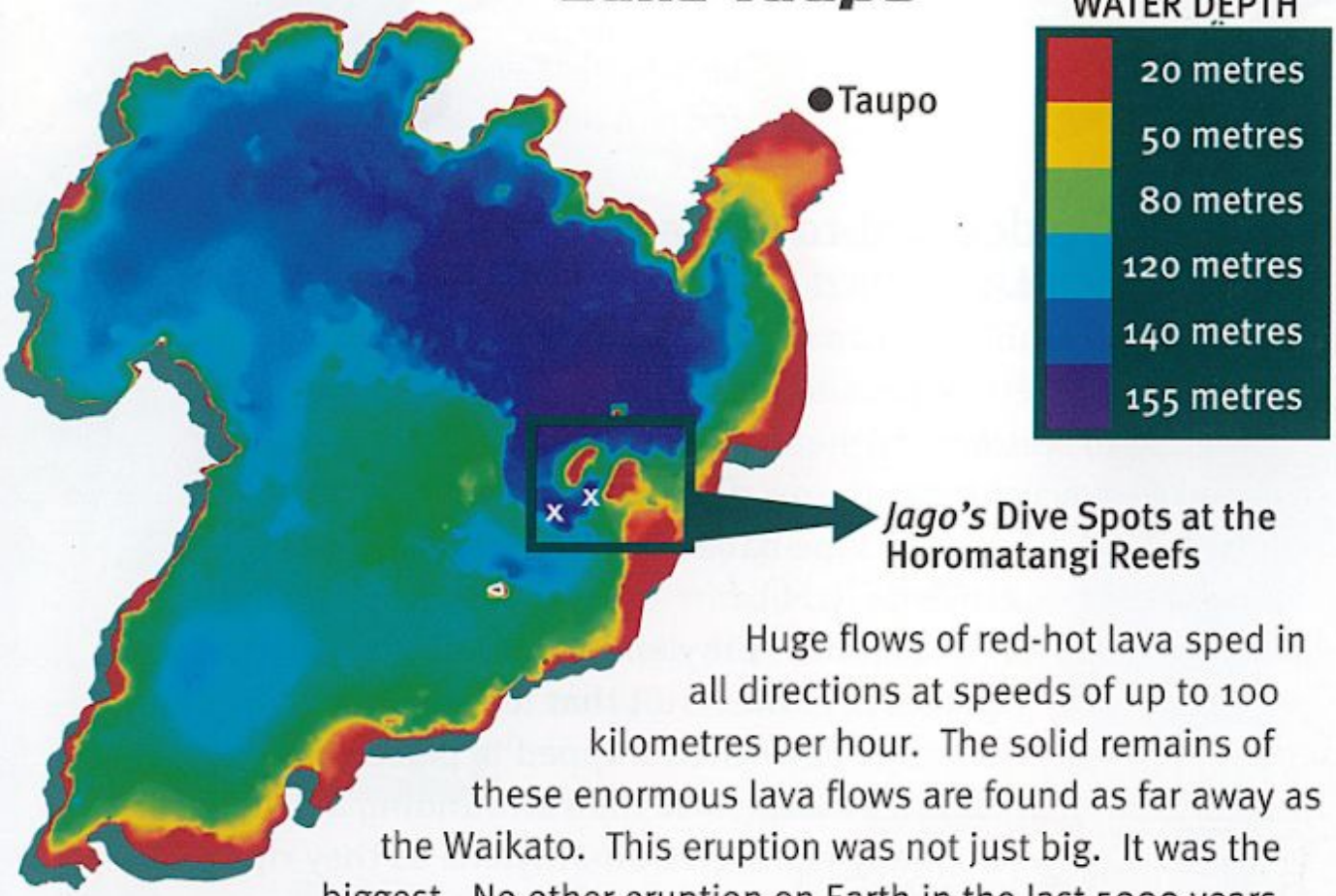


The scientists will use the information they collect to draw a big map of Lake Taupo's geothermal system. Geothermal means heated by the Earth. Hot water and steam from beneath the Earth's surface can be used to generate electricity. For example, at Wairaki, just north of Taupo, there's a geothermal power station. In Rotorua, Māori traditionally used hot springs for cooking and cooled mud for dyeing. Even today, many homes in the area use hot water that comes straight from bore holes in the ground.

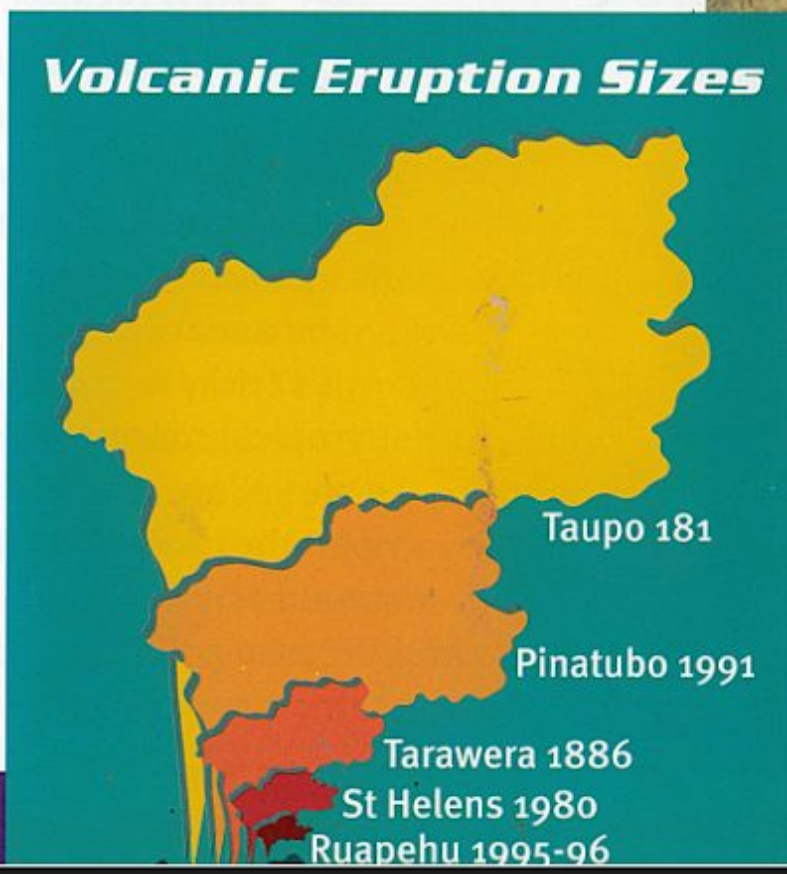
The scientists investigated many hot water and gas vents on the lake bed. It was a very exciting adventure. They gathered lots of interesting data. And it was really thrilling to be in such a weird and unfamiliar environment.

The other exciting thing was the fact that the vents they found are all signs of Lake Taupo's dramatic volcanic history. Taupo is New Zealand's largest lake. In fact, it's a giant volcanic crater that has filled up with fresh water. Taupo last erupted in 181 AD. The eruption was so big that it covered most of the North Island in ash and rocks.

Lake Taupo



Huge flows of red-hot lava sped in all directions at speeds of up to 100 kilometres per hour. The solid remains of these enormous lava flows are found as far away as the Waikato. This eruption was not just big. It was the biggest. No other eruption on Earth in the last 5000 years comes even close in size. The ash thrown into the sky was seen as far away as China. Weeks of strangely bright sunsets were reported in Roman records of the time.

These days, Lake Taupo is calm and beautiful. But the gentle waters lapping at the shore hide an underwater world that's still hissing and bubbling with geothermal energy. The findings from the *Jago* expedition will help us to understand the awesome power of the Earth. Perhaps we'll be able to use this knowledge to harness the energy – or maybe to avoid becoming the victims of another huge eruption.

Volcanic Eruption Sizes